

Ideation Phase

Empathize & Discover — Empathy Map Canvas

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor’s View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Empathy Map Canvas — Rajesh (Small-Scale Farmer)

An empathy map captures knowledge about a user's behaviours and attitudes. Rajesh was chosen for the first empathy map because his constraints (limited data literacy, single-region focus) are the most restrictive of the three personas identified for this project.

Dimension	Content
SAYS	"I always plant what my father planted." / "I wish I knew which crops grow best near my district." / "The market price changes every season and I never know what to expect."
THINKS	"Which crop will give me the best return this Kharif season?" / "Is the weather pattern changing how my region performs?" / "Am I missing out on more profitable crops that others are growing?"
DOES	Relies on word-of-mouth advice from neighboring farmers; follows traditional cropping calendars; makes decisions based on limited local observation and experience
FEELS	Anxious about unpredictable yields; frustrated by lack of reliable, region-specific data; hopeful that technology could help reduce his financial risk
PAINS	No access to historical, region-specific crop performance data; cannot compare seasonal outcomes across years; limited ability to visualize trends or identify high-performing crops
GAINS (What Success Looks Like)	Clear visual insight into which crops perform best in his state and season; confidence in crop selection decisions; reduced financial risk and improved income stability

The pains-to-gains mapping shaped two concrete design decisions: (1) the Farmer's View dashboard uses only three charts — Season-wise Production, Trend of Agricultural Yield, and Cultivated Area by Crop — to avoid overwhelming a first-time user; and (2) every chart uses plain axis labels and a title stating the insight in plain language rather than technical field names.